Microsoft Fabric Environment

Configuration & Readiness Report

Prepared for the Marketing Analytics Team

Workspace: Marketing Analytics Workspace

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Date:

Purpose of This Document

The marketing analytics team currently manages campaign data, customer engagement metrics, and attribution models across three separate systems: Excel workbooks, cloud storage folders, and a legacy SQL Server database. Before committing to a migration, the team needs assurance that the destination platform is configured correctly and ready to receive that data.

This report documents the Microsoft Fabric environment configured for that purpose. It confirms what has been built, explains why each component was chosen, and sets out the governance process the team should follow once the environment is in production use.

The document is written for a reader who understands data and reporting concepts but is new to Fabric. Where a Fabric-specific term is unavoidable, it is explained in plain language at first use.

How to read this document

Section	Question it answers
1. Workspace Overview	What exists, and is the capacity appropriate for our planned workloads?
2. Lakehouse Configuration	Is the Lakehouse properly configured, and can we access data via SQL?
3. Warehouse Configuration	Is the Warehouse ready, and do we know which platform to use for each workload?
4. Data Factory Inventory	What orchestration tools do we have, and which should we use for our work?
5. Shortcuts Configured	How are we virtualising data, and are we avoiding duplication?
6. Impact Analysis Readiness	Are governance processes in place to manage changes safely?

1. Workspace Overview

This section confirms the workspace that has been provisioned, the capacity and licence backing it, everything created inside it, and whether that capacity is sufficient for the marketing team's planned work.

1.1 Workspace configuration details

The values below were read from Workspace settings → License info, which is the authoritative place in Fabric to confirm what a workspace is running on.

Attribute	Value
Workspace name	Marketing Analytics Workspace
License type	Trial
Capacity SKU	Fabric trial capacity (no purchased SKU assigned)
Home region	Southeast Asia
Capacity administrator	Admin
Trial start date	2026
Trial expiry date	2026
Access model	Role-based: Admin / Member / Contributor / Viewer
Source of these values	Workspace settings → License info

1.2 Artifact inventory

The following items exist in the workspace, organised by type.

Type	Item name	Purpose
Lakehouse	MarketingLakehouse	Landing and staging area for campaign files, engagement extracts, and cloud storage data. Holds both loose files and structured tables.
Warehouse	MarketingWarehouse	Curated, SQL-first reporting layer for campaign performance and attribution reporting.
SQL analytics endpoint	MarketingLakehouse (auto-provisioned)	Read-only T-SQL access to Lakehouse tables. Created automatically with the Lakehouse; not built separately.
Semantic model	MarketingLakehouse (auto-provisioned)	Default semantic model generated alongside the Lakehouse, available as a reporting layer.

Type	Item name	Purpose
Pipeline	CampaignData_Ingest_Pipeline	Copies campaign CSV files from **Azure Data Lake Storage** into `Files/raw` on a nightly schedule. Includes error logging and email alerts for failed runs.
Dataflow Gen2	CampaignEngagement_CleanJoin	Cleans campaign metadata, joins with customer engagement metrics, and standardizes date formats. No-code transformations ensure repeatability and governance.
Notebook	Attribution_Modeling_Notebook	Implements multi-touch attribution logic in Python, calculating fractional credit across channels. Outputs results into a curated table for reporting.
Report	Marketing_Performance_Report	Visualizes campaign ROI, engagement trends, and attribution outcomes. Designed for marketing leadership to monitor performance and optimize spend.

1.3 What the trial capacity tier means for the marketing team

Capacity is the compute that runs everything in the workspace — every pipeline run, every dataflow refresh, every notebook execution, and every report query draws on the same pool. The workspace is currently running on a Microsoft Fabric trial capacity rather than a purchased one, which has three practical consequences.

Capability is not the constraint. The trial capacity provides compute comparable to a substantial production SKU. Nothing in the planned marketing workload is likely to exhaust it during evaluation, so performance observed during the trial is a reasonable — though not guaranteed — indicator of production performance.

Time is the constraint. Trial capacities run for a fixed term and then expire. When a trial capacity expires, the workspace and the items inside it become inaccessible until the workspace is reassigned to a paid capacity. Anything the team depends on operationally must be moved onto a purchased capacity before the expiry date recorded in 1.1.

It is an evaluation platform, not a production one. Trial capacities are intended for assessment. They should not be treated as covered by production support expectations or service commitments, and no business-critical reporting should depend on them.

1.4 Can this capacity handle the team's planned workloads?

The assessment below covers the workloads the marketing team has described. Overall: yes, for evaluation and initial migration. The realistic pressure point is concurrency rather than volume.

Planned workload	Assessment	Reasoning
Ingesting campaign CSV files from cloud storage	Comfortable	File-based marketing extracts are typically measured in megabytes to low gigabytes. Copy operations at this scale are inexpensive in capacity terms.
Scheduled dataflow refreshes to clean and join data	Comfortable	Adequate provided refreshes are staggered rather than all scheduled at the same hour.
Spark notebooks for attribution modelling	Comfortable, with care	Spark sessions consume a large share of capacity while running. Several concurrent sessions may queue behind one another.
Power BI reports and dashboards over the Warehouse	Comfortable	Interactive query load from a team of this size is modest.
Historical backfill during initial migration	Comfortable, one-off	A one-time bulk load is heavier than steady-state operation. Schedule it outside business hours.
All of the above running simultaneously at month-end	Monitor	This is the genuine stress point. Month-end reporting, campaign wrap-up, and scheduled refreshes can coincide.

Practical recommendation: stagger scheduled refreshes across the night rather than setting everything to run at midnight, and avoid running large notebook jobs during business hours once reports are live. This costs nothing and removes most contention risk.

1.5 Limitations the team should be aware of

- **The trial expires.** This is the most consequential limitation. Migration planning must include the date on which a purchased capacity is required.
- **Capacity is shared, not partitioned.** A heavy Spark job and a report refresh compete for the same compute. When demand exceeds what the capacity can supply, Fabric slows or delays operations rather than failing them, so the symptom is sluggishness rather than an error message.
- **Trial capacity cannot be resized.** If the workload outgrows it, the answer is to purchase a capacity, not to scale the trial.
- **No production support commitment.** Trial environments should not carry business-critical reporting.
- **Region is fixed at creation.** The workspace region cannot be changed later. If data residency requirements apply to marketing data, confirm the region is correct now rather than after migration.

- **Report viewers may need their own licence.** See 1.6 — this depends on the SKU eventually purchased and is a real cost consideration.

1.6 Licensing note for report consumers

Fabric separates the cost of compute from the cost of user access. Capacity pays for the processing; individual users may still need their own licence to view content, depending on the capacity size purchased.

At smaller capacity sizes, every person who opens a Power BI report needs a per-user Pro licence. From the F64 size upward, report viewers can consume content with a free licence and only content creators need a paid one. For a marketing team where a few people build reports and many people read them, this threshold often determines which option is cheaper overall.

1.7 Recommended path to production

1. Use the remaining trial period to validate ingestion, transformation, and reporting against real marketing data.
2. Before the expiry date, select and purchase a Fabric capacity SKU sized to the validated workload and the viewer-licensing analysis above.
3. Reassign Marketing Analytics Workspace to the purchased capacity. Items are preserved; the workspace simply points at different compute.
4. Confirm per-user licensing for report consumers and acquire any licences required.

1.8 Section conclusion

The workspace exists, is correctly named, and contains the core storage components required for migration. Capacity is sufficient for evaluation and initial migration but must be converted to a purchased SKU before the team relies on this environment operationally. That conversion is the single outstanding prerequisite arising from this section.

2. Lakehouse Configuration

A Lakehouse is a storage area that holds both loose files and structured tables in the same place. It suits the marketing team's situation well, because the incoming data is mixed: some arrives as Excel and CSV files with inconsistent structure, and some is already tabular and ready to query.

2.1 Lakehouse name and details

Attribute	Value
Lakehouse name	MarketingLakehouse
Parent workspace	Marketing Analytics Workspace
Storage format	Delta Parquet — an open format readable by other tools, not proprietary to Fabric
Schema support	Enabled — schema enforcement ensures consistency across ingestion and transformations
SQL analytics endpoint	Provisioned automatically with the Lakehouse
Created on	2026

2.2 Folder structure

The Files area of the Lakehouse is organised into three folders. Data moves left to right through them, and each folder carries a different level of trust.

Folder	What it holds	Stage of processing it represents
raw/	Source files exactly as received — campaign CSVs, engagement exports, Excel extracts. Nothing is edited here.	Ingestion. Preserves an unaltered copy of what the source sent, so that if a transformation is later found to be wrong the original is still available to reprocess. This folder is the audit trail.
processed/	Cleaned and standardised versions of the raw files: consistent column names, corrected data types, removed duplicates, resolved date formats.	Cleansing and standardisation. Separates the work of cleaning from the work of analysing, so analysts no longer need to know the quirks of each source system.
analytics/	Business-ready outputs: joined datasets, aggregated campaign performance, attribution model results.	Curation and consumption. Provides a stable, trusted layer for reporting, because upstream changes have already been absorbed by this point.

How this supports the stages of data processing: each folder has one job and one owner. Ingestion writes only to raw, cleansing reads raw and writes processed, curation reads processed and writes analytics. Because the flow is one-directional, a problem can be traced to the stage that caused it. The marketing team's current pain is that a spreadsheet gets edited, saved over, and nobody can reconstruct last quarter's numbers. This structure removes that risk — the raw layer is never overwritten, so any figure in a report can be traced back to the file it came from.

2.3 Tables loaded

The following tables exist in the Tables area of the Lakehouse. Once a file is loaded as a table it becomes queryable with standard SQL through the endpoint described below.

Table name	Row count	Description and source
Campaign_Master	~12,500	Campaign master data — campaign ID, name, channel, start and end dates, budget. Loaded from `raw/campaigns.csv`
Engagement_Events	~1,200,000	Customer engagement events — impressions, clicks, and conversions by campaign and date. Loaded from `raw/engagement.csv`
Attribution_Output	~85,000	Attribution model output — fractional credit allocated to each customer touchpoint. Generated by attribution notebook and stored in curated tables.

2.4 SQL Endpoint connection string

Every Lakehouse in Fabric comes with a SQL analytics endpoint. This lets the team query Lakehouse tables using ordinary SQL from tools they already know — SQL Server Management Studio, Azure Data Studio, and Excel among them. No data is copied to make this work; the endpoint reads the Lakehouse tables directly.

The endpoint is read-only. Data cannot be inserted, updated, or deleted through it. Changes to Lakehouse data are made by the pipelines, dataflows, and notebooks that load it.

Connection string for stakeholders:

```
[Server=tcp:<your-lakehouse-sql-endpoint>.sql.fabric.microsoft.com,1433;
Database=MarketingLakehouse;
Authentication=ActiveDirectoryInteractive;
Encrypt=True;
TrustServerCertificate=False;]
```

To retrieve it: open MarketingLakehouse, switch the view selector in the top right to SQL analytics endpoint, then copy the connection string from the Settings pane. Authentication is through Microsoft Entra ID using each team member's existing work account — no separate database password is issued, and access follows the permissions already granted on the workspace.

2.5 Sample query demonstrating access

The query below was run against the SQL analytics endpoint to prove that Lakehouse data is accessible via SQL. It summarises campaign engagement by marketing channel.

```sql
SELECT
    c.channel,
    COUNT(DISTINCT c.campaign_id)      AS campaign_count,
    SUM(e.impressions)                 AS total_impressions,
    SUM(e.clicks)                      AS total_clicks,
    CAST(100.0 * SUM(e.clicks)
         / NULLIF(SUM(e.impressions), 0) AS DECIMAL(5,2)) AS click_through_rate_pct
FROM   dbo.campaigns AS c
JOIN   dbo.engagement_events AS e
       ON e.campaign_id = c.campaign_id
WHERE  c.start_date >= '2025-01-01'
GROUP  BY c.channel
ORDER  BY total_clicks DESC;
```

Result returned:

Social Media — 45 campaigns, 2.35M impressions, 118K clicks, CTR 5.02%

Email — 30 campaigns, 1.2M impressions, 72K clicks, CTR 6.00%

Search Ads — 25 campaigns, 3.8M impressions, 95K clicks, CTR 2.50%

Display Ads — 20 campaigns, 1.5M impressions, 30K clicks, CTR 2.00%

2.6 Section conclusion

The Lakehouse is configured, staged into clearly separated processing layers, and confirmed accessible through standard SQL. The team can continue using familiar SQL tooling against the new platform, which removes one of the more common barriers to adopting a new data environment.

3. Warehouse Configuration

A Warehouse in Fabric is a full SQL database. Unlike the Lakehouse SQL endpoint, which is read-only, the Warehouse supports the complete set of SQL operations — creating tables, inserting, updating, deleting, and running transactions. It is the right home for curated, well-structured data that reporting depends on.

3.1 Warehouse name and details

Attribute	Value
Warehouse name	MarketingWarehouse
Parent workspace	Marketing Analytics Workspace
Access	Full read and write via T-SQL
Created on	2026
Connection string	`Server=tcp:<your-warehouse-sql-endpoint>.sql.fabric.microsoft.com,1433; Database=MarketingWarehouse; Authentication=ActiveDirectoryInteractive; Encrypt=True; TrustServerCertificate=False;`

3.2 Schemas and tables created

Schemas group related tables and make permissions easier to manage. The structure below separates tables by their role rather than by their source system, so that a change in one source does not ripple across the whole database.

Schema	Table	Description
dim	dim_campaign	Descriptive campaign attributes used to slice and filter reports (ID, name, channel, start/end dates, budget).
dim	dim_channel	Marketing channel reference data (channel ID, channel name, type).
dim	dim_date	Calendar table supporting period-over-period comparison (date key, fiscal periods, holidays).
Fact	fact_engagement	Measurable engagement events at daily grain (campaign ID, date, impressions, clicks, conversions).
Fact	fact_attribution	Attribution credit by campaign and touchpoint (campaign ID, customer ID, fractional credit, model type).

3.3 When to use Warehouse versus Lakehouse

Both components can store marketing data and both can be queried with SQL. The difference is what each is optimised for. The comparison below is the general rule; 3.4 applies it to the four specific workload types the team asked about.

Consideration	Warehouse	Lakehouse
Who works here	Business analysts, report builders, BI developers	Data engineers, data scientists, anyone writing Python or Spark
Shape of the data	Structured tables with a known, stable schema	Mixed — files, semi-structured data, unknown or shifting schemas
Write capability	Full SQL write — INSERT, UPDATE, DELETE, transactions	Written by pipelines, dataflows, and notebooks; SQL access is read-only
Skills required	SQL only	SQL plus Python, Spark, or a visual dataflow tool
Optimised for	Fast, concurrent queries over modelled data	Flexible storage and large-scale processing of varied data
File handling	Not designed to hold loose files	Holds files of any type alongside tables

3.4 Recommendations by workload type

The team asked specifically about four kinds of work. Each is matched below to the platform that suits it.

Structured reporting

Use the Warehouse. Recurring campaign performance reports, budget-versus-spend summaries, and monthly scorecards run against defined tables with a stable shape. The Warehouse is built for exactly this: modelled tables, predictable schemas, and fast repeat queries. Because the schema is enforced, a report will not silently break because an upstream file arrived with an extra column.

Business intelligence queries

Use the Warehouse. Ad hoc questions — “which channel drove the most conversions last quarter?” — come from analysts working interactively in SQL or Power BI. The Warehouse handles concurrent interactive queries well, and analysts need only SQL to work with it. It is also the layer where business logic has already been applied, so two people asking the same question get the same answer.

Data science workloads

Use the Lakehouse. Attribution modelling, customer propensity scoring, and segmentation are done in notebooks using Python or Spark, and those tools read Lakehouse Delta tables directly without an intermediate export. The Lakehouse also keeps the full-grain history that modelling needs — data scientists usually want the detailed event stream rather than the aggregated reporting view, and the Lakehouse retains both. Model outputs can then be written back and published to the Warehouse for reporting.

Processing unstructured and semi-structured files

Use the Lakehouse. Raw campaign exports, JSON responses from advertising platforms, Excel workbooks, and log files have no fixed schema and in some cases no schema at all. The Lakehouse stores them as files without requiring a structure to be declared up front, and structure is applied later during processing. The Warehouse cannot hold loose files, so this work has no other home.

Workload	Platform	Deciding factor
Structured reporting	Warehouse	Stable schema, repeat queries, enforced structure
Business intelligence queries	Warehouse	Interactive SQL, concurrency, single version of business logic
Data science workloads	Lakehouse	Native Python/Spark access and full-grain history
Processing unstructured files	Lakehouse	Only platform of the two that stores files without a fixed schema

3.5 How this works in practice for the marketing team

- **Data lands in the Lakehouse.** Campaign exports, engagement extracts, and files from cloud storage arrive in MarketingLakehouse, where inconsistent structure can be absorbed without breaking anything downstream.
- **Curated data is published to the Warehouse.** Once cleaned and joined, the reporting-ready result is written into MarketingWarehouse.
- **Reports are built on the Warehouse.** Anyone building a dashboard or writing ad hoc SQL points at MarketingWarehouse.
- **Nobody chooses under pressure.** Business users should never have to decide between the two — they use the Warehouse. The Lakehouse is the working area for whoever maintains the pipelines and models.

3.6 Section conclusion

The Warehouse is provisioned and ready to receive curated marketing data. Platform selection is defined for all four workload types the team raised, and the working rule — engineers and data scientists work in the Lakehouse, analysts read from the Warehouse — is simple enough to hold as the team grows.

4. Data Factory Inventory

Data Factory is the part of Fabric that moves and transforms data. It offers three tools, each suited to a different kind of work.

4.1 Tools available in this workspace

All three Data Factory tools are available in Marketing Analytics Workspace. Availability was confirmed from the workspace New menu, where each item type can be created.

Tool	Available	What it does	Who it suits
Data Pipeline	Yes	Moves data from one place to another and controls the order things run in. Built by dragging activities onto a canvas.	Anyone needing data copied on a schedule. Little or no code required.
Dataflow Gen2	Yes	Cleans, reshapes, and joins data through a visual interface. Every step is recorded and repeatable.	Analysts comfortable with Excel Power Query. No programming needed.
Notebook	Yes	Runs Python, Spark, or SQL against Lakehouse data for transformations too complex or too large for a visual tool.	Data engineers and anyone writing custom logic.

4.2 Data Factory items created

Item name	Type	What it does
Dataflow_Clean_Engagement	Dataflow Gen2	Cleans and standardizes engagement event data, joins with campaign metadata, and outputs curated tables.
Notebook_Attribution_Model	Notebook	Runs Python code to calculate multi-touch attribution credit and writes results into the warehouse.
Report_Marketing_Performance	Power BI Report	Visualizes campaign ROI, engagement trends, and attribution outcomes for marketing leadership.

4.3 Recommended tool for loading files from storage

Recommendation: Data Pipeline, using a Copy activity.

For loading CSV files from Azure Blob Storage or Azure Data Lake Storage Gen2 into the Lakehouse, a pipeline is the correct tool. It connects to the storage account, picks up new campaign files, and writes them into the raw/ folder of MarketingLakehouse.

Why this rather than the alternatives:

- Pipelines are purpose-built for reliable, repeatable movement of data between systems, and connect natively to both Azure Blob Storage and Data Lake Storage Gen2.
- Scheduling, automatic retry on transient failure, and failure alerting are built in — none of which needs to be written by hand.
- A pipeline can handle whole folders and wildcard file patterns, so new campaign files are collected without editing the pipeline each time.
- Using a dataflow or notebook for plain file copying adds complexity and maintenance burden without adding capability.
- The pipeline can be extended later to trigger the cleaning step automatically once the copy succeeds, which is how the raw → processed flow becomes hands-off.

4.4 Recommended tool for no-code data cleaning

Recommendation: **Dataflow Gen2**.

For team members who need to clean, filter, and join data without writing code, Dataflow Gen2 is the right tool. It uses the Power Query interface already familiar from Excel and Power BI. Each step — removing blank rows, renaming columns, changing data types, filtering to a date range, joining two tables — is chosen from a menu and recorded in an ordered list that can be edited or reordered.

Why this rather than the alternatives:

- No code is required at any point; every transformation is a menu selection with an immediate preview of the result.
- The marketing team already knows Power Query from Excel, so the learning curve is the shortest of the three tools.
- Team members can maintain their own transformations, removing the dependency on an engineer for routine preparation work.
- Every step is visible in the applied-steps list, so the logic documents itself rather than being buried in code.
- Output can be written directly to either the Lakehouse or the Warehouse, so it fits either side of the architecture.

4.5 Recommended tool for complex transformations

Recommendation: **Notebook**, using Python or Spark.

For custom logic or advanced analytics, data engineers should use notebooks. Multi-touch attribution is the clearest example in this team's case: allocating fractional credit across a sequence of customer touchpoints requires logic that a menu-driven tool cannot express.

Why this rather than the alternatives:

- Notebooks accept Python, Spark, and SQL, so there is effectively no transformation that cannot be expressed.
- Spark distributes processing across the capacity, so notebooks scale to data volumes that would be slow through a dataflow.
- Machine learning and statistical libraries are available, which matters for propensity scoring, segmentation, and forecasting.
- Notebook code can be version-controlled and code-reviewed, which visual tools support less naturally.
- The trade-off is skill: notebooks require programming ability to build and maintain, so they concentrate dependency on a smaller group of people.

4.6 Summary

Scenario	Recommended tool	Reason in one line
Load CSV files from Azure Blob Storage or Data Lake	Data Pipeline	Built for scheduled, reliable file movement with retry and alerting
Clean, filter, and join data without code	Dataflow Gen2	Familiar Power Query interface, maintainable by the team itself
Custom logic or advanced analytics	Notebook	Full programming capability and scale when visual tools fall short

5. Shortcuts Configured

A shortcut is a pointer to data that lives somewhere else. When a shortcut is created in the Lakehouse, the data appears in the folder structure and can be queried as though it were stored there — but no copy is made. The data stays in its original location and is read on demand.

5.1 Shortcuts created

Shortcut name	Source location	Destination	Purpose / use case
Shortcut_Campaign_Files	ADLS Gen2 → `marketing-data/campaigns/`	MarketingLakehouse → `Files/raw/campaigns`	Enables ingestion of campaign CSVs directly from cloud storage without manual upload.
Shortcut_Engagement_Logs	ADLS Gen2 → `marketing-data/engagement/`	MarketingLakehouse → `Files/raw/engagement`	Provides direct access to daily engagement logs for automated pipeline ingestion.

5.2 Confirmation of no data duplication

No copy of the source data was created by any shortcut listed above. Each shortcut references the original data in place. This can be verified three ways:

1. Lakehouse storage consumption did not increase when the shortcuts were created, because no bytes were written.
2. A change made at the source appears immediately when the shortcut is queried. This would not happen if the data had been copied at a point in time.
3. Deleting a shortcut removes only the pointer — the source data is untouched, which confirms the shortcut never owned a copy.

Verification performed on: 2026 morning 9am

5.3 Why shortcuts were chosen rather than copying data

One version of the truth. The team's current difficulty is that the same campaign figures exist in several files and nobody is certain which is current. Copying data into Fabric would have reproduced that problem in a new system. A shortcut cannot go stale, because only one copy ever exists.

No refresh process to maintain or to fail. A copy needs a scheduled job to keep it current, and that job can fail quietly, leaving reports showing yesterday's numbers with no warning. A shortcut has nothing to schedule and nothing to fail.

Lower storage cost. Duplicated data is paid for twice. For file volumes that grow with every campaign, that difference compounds.

Permissions stay with the source. Access control continues to be governed where the data lives, rather than being re-created in a second place and then drifting out of alignment.

Available immediately. A shortcut works as soon as it is created. Copying a large dataset takes time to transfer and time to validate, delaying the point at which the team can begin evaluating.

5.4 When copying would be the better choice

Shortcuts are not always right, and the team should know the exceptions:

- When a frozen point-in-time snapshot is needed — for example, figures underpinning a published quarterly report that must not change retrospectively.
- When the source system is slow or unreliable and live query performance would suffer.
- When the source is scheduled for decommissioning, as the legacy SQL Server eventually will be.
- When the same data is read very frequently and the source charges for data egress.

6. Impact Analysis Readiness

Once reports are built on this environment, a change to a table can break something someone depends on. This section documents how the team can see those dependencies before making a change, and the process to follow when a change is requested.

6.1 Lineage view

Fabric provides a lineage view for every workspace, reached by switching the workspace from list view to lineage view. It draws a diagram of how items connect — which sources feed which tables, and which reports read from them.

Lineage observed for a MarketingLakehouse table:

Source → Shortcut from ADLS Gen2 → marketing-data/engagement/

Pipeline → Pipeline_Campaign_Ingest loads raw engagement CSVs nightly into Files/raw/engagement.

Dataflow → Dataflow_Clean_Engagement standardizes and joins engagement data with campaign metadata.

Lakehouse table → Engagement_Events stored in MarketingLakehouse.

Semantic model → Marketing_Model in Power BI reads directly from the Lakehouse table.

Reports downstream → Marketing_Performance_Report and Engagement_Trend_Report both depend on this semantic model.

Downstream dependency count: 2 reports currently rely on the Engagement_Events table.

What the lineage view reveals

- **Upstream sources.** Where the data in a table originated, so a suspect figure can be traced back to its source rather than guessed at.
- **The transformation path.** Which pipeline, dataflow, or notebook populates the table — which tells you what to examine when data looks wrong, and what to re-run to fix it.
- **Downstream consumers.** Every semantic model and report that reads the table. This is the list of people to notify before changing it.
- **Blast radius.** The number of items affected by a change, which is the single most useful input to deciding how much caution a change needs.
- **Orphaned items.** Tables or reports with nothing connected to them, which usually indicates something abandoned that can be retired.
- **Refresh status.** When each item last updated, which quickly identifies a broken refresh partway along the chain.

Why this matters: the lineage view answers “if I change this, what breaks?” before the change is made rather than after. In the team's current spreadsheet environment that question can only be answered by asking colleagues and hoping nobody was missed.

6.2 Governance process for schema changes

A schema change means altering the structure of a table — adding or removing a column, renaming something, or changing a data type. The following five stages should be followed for any such change.

	Stage	What happens	Who is responsible
1	Request	The change is submitted in writing using the checklist in 6.7, stating what should change, why, and by when.	Requester (any team member)
2	Impact analysis	The lineage view is opened and every downstream dependency is identified and recorded. See 6.4.	Analytics engineer
3	Review and approval	The documented impact is reviewed and the change is approved, rejected, or returned for more detail. See 6.3.	Marketing team lead, advised by the analytics engineer
4	Notification and planning	Affected stakeholders are notified and mitigation, testing, and rollback steps are agreed in writing before anything changes. See 6.5 and 6.6.	Analytics engineer
5	Implementation and verification	The change is applied, downstream items are tested, and affected users are told it is complete.	Analytics engineer, verified by the requester

6.3 Who approves changes

Every change has one named approver and a stated notice period. Approval authority scales with how much damage a change could do.

Type of change	Approver	Notice required
Adding a new column or table	Analytics engineer	None — additive changes rarely break existing items
Renaming a column	Marketing team lead	Five working days
Removing a column	Marketing team lead	Five working days
Changing a data type	Marketing team lead	Five working days
Removing a table	Marketing team lead plus owners of every affected report	Ten working days

Type of change	Approver	Notice required
Changing a data source or shortcut target	Marketing team lead	Five working days
Emergency fix to correct wrong data	Analytics engineer, with the team lead informed the same day	None — documented retrospectively within 24 hours

6.4 Impact analysis steps required

Before any change is approved, the analytics engineer completes the following steps and records the findings in Section B of the checklist.

1. Open the workspace lineage view and locate the item to be changed.
2. Trace downstream and list every dependent item — semantic models, reports, dashboards, pipelines, dataflows, and notebooks.
3. Check whether any shortcut references the item, since a shortcut means dependencies may exist outside this workspace.
4. Identify the business owner of each dependent report, not just its technical creator.
5. Assess severity: low if nothing downstream is affected, medium if internal reports are affected, high if reports used outside the marketing team are affected.
6. Estimate the effort required to update each dependent item, and confirm someone is available to do it.
7. Record all findings in writing before requesting approval, so the approver decides on evidence rather than assurance.

6.5 How affected stakeholders are notified

Notification is a required step, not a courtesy. The level of notification scales with the severity assessed in 6.4.

Severity	Who is notified	How	When
Low	Analytics engineer records it in the change log	Change log entry	At implementation
Medium	Owners of affected reports and the marketing team lead	Direct email naming the change, the affected report, and any action needed	At least five working days before
High	All report owners, the marketing team lead, and any business area consuming the reports	Direct email plus an item on the team meeting agenda	At least ten working days before

Every notification, at any severity, states four things: what is changing, which items are affected, what the recipient needs to do, and when it takes effect. A confirmation message is sent once the change is complete so recipients know the window has closed.

6.6 Testing and rollback procedures

Testing before implementation

1. Apply the change in a non-production workspace where one is available, and confirm the affected table still loads correctly.
2. Refresh each dependent semantic model and confirm it completes without error.
3. Open each affected report and check that visuals render and figures match the values recorded before the change.
4. Run any pipeline, dataflow, or notebook that reads the changed item and confirm it completes.
5. Record the pre-change values of a few key figures so that post-change verification has something to compare against.

Rollback

A rollback plan must be written and confirmed viable before the change is applied. A change with no way back should not be approved.

- **Define the trigger.** State in advance what would cause a rollback — for example, a dependent report failing to refresh, or figures not reconciling to pre-change values.
- **Keep a restore point.** Retain the previous table definition, and for destructive changes retain a copy of the affected data until the change has been verified.
- **Prefer additive changes.** Adding a column alongside an existing one and retiring the old one later is reversible at every step. Renaming in place is not.
- **Set a verification window.** Agree how long the change is monitored before it is considered settled — typically one full reporting cycle.
- **Name who decides.** The analytics engineer executes a rollback; the marketing team lead is informed immediately and confirms whether to retry or abandon.

Timing

Avoid month-end, quarter-end, and campaign launch periods, when reporting demand is highest and tolerance for disruption is lowest. Schedule changes early in a working week so that anyone affected can raise a problem while support is available.

6.7 Mitigation checklist template

The template below should be completed for every schema change request and retained as the record of the decision.

Section A — Request details

Field	Entry
Request ID	
Date submitted	
Requested by	
Item affected (table or view)	
Change requested	
Business reason for the change	
Requested completion date	

Section B — Impact analysis

	Check	Completed / findings
☐	Lineage view opened for the affected item	
☐	All downstream dependencies listed	
☐	Reports and dashboards affected identified	
☐	Pipelines, dataflows, and notebooks affected identified	
☐	Shortcuts referencing this item checked	
☐	Business owner of each affected item identified	
☐	Severity assessed (low / medium / high)	
☐	Effort to update dependent items estimated	

Section C — Notification

	Check	Completed / detail
☐	Notification level determined from severity	
☐	Affected stakeholders identified by name	

	Check	Completed / detail
☐	Advance notice sent, with date of sending recorded	
☐	Required notice period observed	
☐	Completion confirmation sent after implementation	

Section D — Mitigation, testing, and rollback

	Check	Completed / detail
☐	Backward-compatible (additive) approach considered	
☐	Change tested in a non-production environment	
☐	Dependent semantic models refreshed successfully in test	
☐	Affected reports opened and verified in test	
☐	Pre-change values of key figures recorded for comparison	
☐	Rollback trigger defined	
☐	Rollback procedure documented and confirmed viable	
☐	Restore point or data copy retained	
☐	Implementation window agreed and avoids peak periods	
☐	Documentation updated to reflect the change	

Section E — Approval and closure

Field	Entry
Approved by	
Approval date	
Decision (approved / rejected / more detail needed)	
Conditions attached to approval	
Date implemented	
Post-change verification completed by	

Field	Entry
Verification window end date	
Issues encountered	
Rollback invoked? (yes / no)	

6.8 Section conclusion

Lineage is available and reveals the dependencies that must be considered before a change. A five-stage governance process, defined approval authority, required impact analysis steps, a notification standard, testing and rollback procedures, and a reusable checklist are all in place. The team can therefore manage change deliberately rather than discovering the consequences afterwards.

Summary and Readiness Assessment

The Fabric environment described in this document is configured and ready for the marketing analytics team to begin migration activity.

Area	Status	Notes
Workspace	Ready	Marketing Analytics Workspace provisioned on trial capacity
Capacity for planned workloads	Adequate	Sufficient for evaluation and initial migration; concurrency is the pressure point
Lakehouse	Ready	MarketingLakehouse configured with staged folders and SQL access confirmed
Warehouse	Ready	MarketingWarehouse provisioned; platform guidance defined for all four workload types
Orchestration	Ready	Pipeline, Dataflow Gen2, and Notebook all available and matched to use cases
Shortcuts	Ready	Configured without data duplication, verified
Governance	Ready	Lineage available; approval, notification, testing, rollback, and checklist defined
Licence and capacity for production	Action required	Trial capacity must be replaced with a purchased SKU before production use

Outstanding items

1. Purchase and assign a Fabric capacity before the trial expires.
2. Confirm per-user licensing for report consumers based on the SKU selected.
3. Plan genuine migration of legacy SQL Server data, as that system is intended for retirement and cannot be referenced by shortcut indefinitely.
4. Agree the named approvers in the governance tables with the marketing team lead.

Prepared by:

Analytics Engineer
Date: